

System Identification and Control of a Propofol Anesthesia Delivery System

MECH 436 Final Project Presentation

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1. Motivation and Problem Statement

Clinical control objective

Maintain a safe depth of hypnosis by adjusting the propofol infusion rate using feedback from an EEG-based monitor.

- Input: propofol infusion rate.
- Output: measurable depth-of-hypnosis indicator.
- Challenge: patient dynamics are delayed, high order, noisy, and uncertain.

Why delay matters

$$G(s) = Ke^{-T_d s} \frac{(s + z_1)(s + z_2)}{(s + p_1)(s + p_2)(s + p_3)(s + p_4)}$$

$$\angle e^{-j\omega T_d} = -\omega T_d$$

Delay adds phase lag, reduces phase margin, and can destabilize aggressive controllers.

2. System Identification Procedure

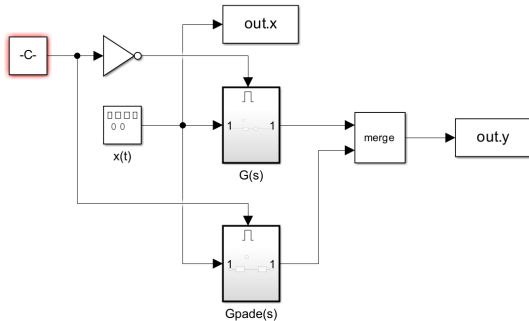


Figure 1: Black-box frequency sweep model.

Sinusoidal inputs were applied over logarithmically spaced frequencies:

$$x(t) = \sin(\omega t), \quad y_{ss}(t) = Y \sin(\omega t + \phi)$$

The steady-state output was fitted as

$$y_{ss}(t) = b_1 \sin(\omega t) + b_2 \cos(\omega t) + c$$

using least squares:

$$b = A^+ y_{ss}$$

Then

$$|G(j\omega)| = \sqrt{b_1^2 + b_2^2}, \quad \angle G(j\omega) = \text{atan2}(b_2, b_1)$$

3. Identification Results: Model, Delay, and Padé Approximation

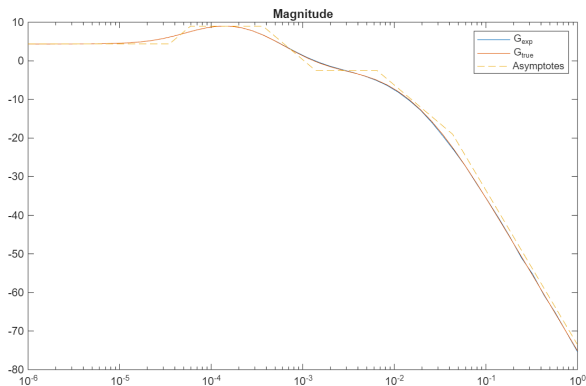


Figure 2: Magnitude Bode plot and asymptotic fit.

The identified no-delay model was

$$G_{id}(s) = \hat{K} \frac{\prod(s + \hat{z}_i)}{\prod(s + \hat{p}_i)}$$

with

$$\hat{K} = 2.08 \times 10^{-4}$$

Phase drift revealed an unmodeled delay:

$$\hat{T}_d = -\frac{d}{d\omega} (\Delta \angle G(j\omega)) \approx 10.36 \text{ s}$$

For controller design, the delay was approximated by

$$e^{-T_d s} \approx \frac{1 - \frac{T_d}{2}s}{1 + \frac{T_d}{2}s}$$

4. Noise Effects in Identification

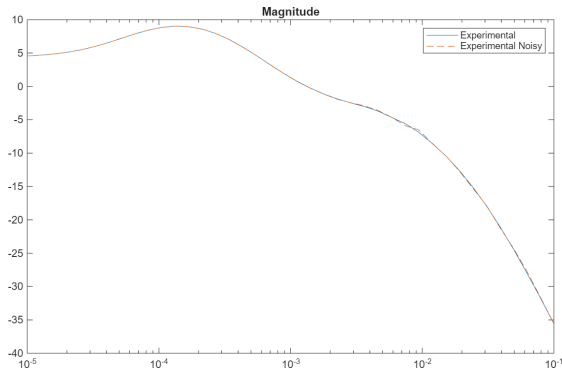


Figure 3: Magnitude with and without noise.

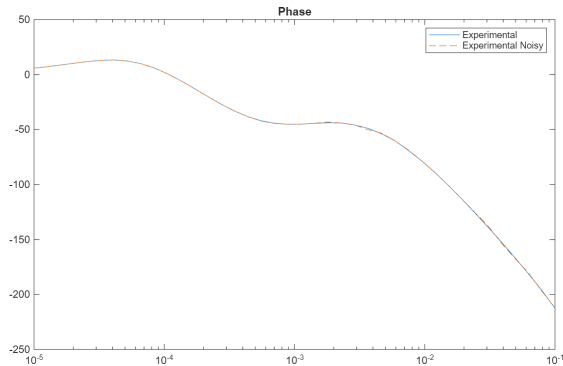


Figure 4: Phase with and without noise.

Main observation

Zero-mean noise is partly rejected by the least-squares harmonic fit, while drift and noise near the excitation frequency can bias the extracted frequency response.

5. Translating Specifications into Design Targets

Project requirements

$$T_s \leq 1200 \text{ s}, \quad M_p \leq 3\%, \quad e_{ss} \leq 0.02$$

Ideal response

$$T_{ideal}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Second-order benchmark

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\zeta = 0.7448$$

$$\omega_n = \frac{4}{\zeta T_s} = 4.475453 \times 10^{-3} \text{ rad/s}$$

Controller families

$$C_{LL}(s) = K_c \frac{s + z_{lead}}{s + p_{lead}} \frac{s + z_{lag}}{s + p_{lag}}$$

$$C_{PID}(s) = K_{PID} \frac{(s + a)(s + b)}{s(T_f s + 1)}$$

6. Controller Design Methodology

Root-locus design

- Convert specifications to desired dominant poles.
- Use angle condition to place lead zero/pole.
- Use magnitude condition to compute gain.
- Add lag network for steady-state error.
- PID zeros selected near dominant dynamics; derivative filter added for realizability.

Frequency-response design

- Select desired crossover frequency and phase margin.
- Lead network supplies missing phase.
- Lag network increases low-frequency gain.
- PID zero/filter locations selected from phase requirement at crossover.
- Verify margins, delay robustness, and control effort.

Verification principle

Controllers were designed using the rational Padé model, then verified on the original delayed plant.

7. MATLAB Verification: Comparison of All Designed Controllers

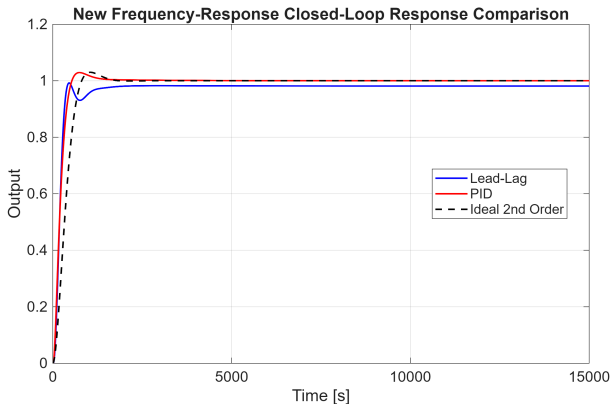


Figure 5: Frequency-response controllers compared with the ideal second-order response.

Table 1: Nominal performance comparison of all designed controllers.

Controller	T_s (s)	M_p (%)	e_{ss}
RL Lead-Lag	1025.02	2.21	0.01855
RL PID	582.07	1.97	0.00000
FR Lead-Lag	1073.74	1.10	0.01855
FR PID	975.31	2.91	0.00000

Main comparison

All four controllers satisfy the nominal requirements:

$$T_s \leq 1200 \text{ s}, \quad M_p \leq 3\%, \quad e_{ss} \leq 0.02.$$

PID controllers eliminate steady-state error, while lead-lag controllers give smoother and more conservative responses.

8. Margins, Delay Robustness, and Control Effort

New Frequency-Response Lead-Lag Open-Loop Margin

Gm = 18 dB (at 0.0216 rad/s), Pm = 67.4 deg (at 0.00553 rad/s)

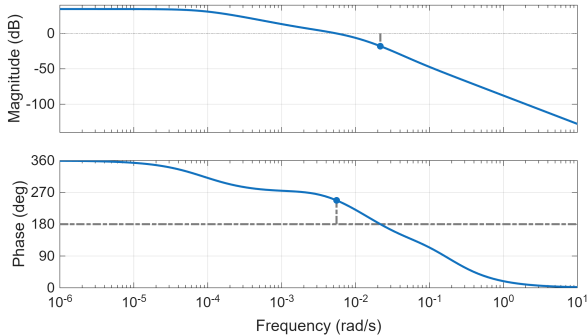


Figure 6: Lead-lag open-loop margins.

Table 2: Frequency-response design robustness.

Controller	GM (dB)	PM (deg)
Lead-Lag	17.97	67.42
PID	25.73	70.00

Table 3: Delay robustness.

Controller	$1.0 T_d$	$1.4 T_d$	$1.6 T_d$
LL M_p	1.10%	2.32%	2.96%
PID M_p	2.91%	3.20%	3.37%

Lead-lag remains below the overshoot limit under tested delay increase; PID remains stable but crosses the 3% bound.

9. Simulink Validation with Saturation and Sensor Noise

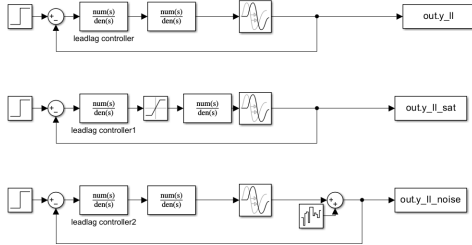
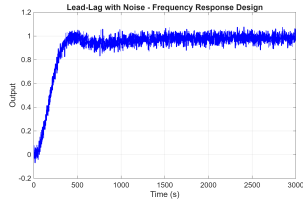
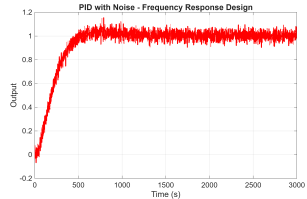


Figure 7: Closed-loop Simulink model with delay, saturation, and noise.



(a) LL + noise

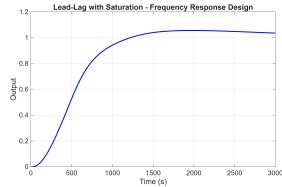


(b) PID + noise

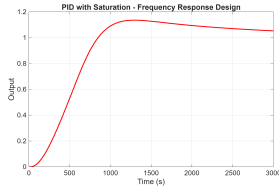
Figure 8: Noise sensitivity in Simulink.

PID tracks better nominally, but derivative action increases sensitivity to high-frequency measurement noise.

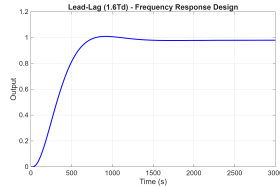
10. Simulink Practical Effects: Saturation and Delay Variation



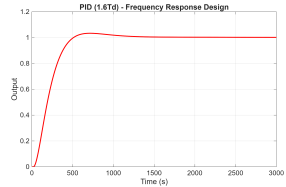
(a) LL sat.



(b) PID sat.



(a) LL, $1.6T_d$



(b) PID, $1.6T_d$

Figure 9: Actuator saturation.

Figure 10: Delay robustness.

Implementation takeaway

Saturation penalizes the PID more strongly because it demands larger control effort; lead-lag is more forgiving under actuator limits.

11. Final Conclusions

Main lessons

- The black-box plant can be identified from sinusoidal frequency sweeps.
- Linear phase drift reveals transport delay.
- Padé approximation enables classical controller design.
- MATLAB and Simulink trends agree, but Simulink exposes practical limitations.

Controller choice

- **PID:** faster response and zero steady-state error.
- **Lead-lag:** smoother, lower control effort, more conservative under delay, noise, and saturation.

Final takeaway

For anesthesia delivery, nominal tracking must be balanced against robustness and actuator limitations.